

Physical Chemistry (I)

Lecture 10

Entropy



Last class

Entropy

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

- Entropy change

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

- Molar entropy change $\Delta S_{\text{m}} = \Delta S/n$



Heat Transfer to Surroundings

Consider an infinitesimal transfer of heat

$$\delta q = -\delta q_{\text{sur}}$$

The surroundings are extremely large, so

$$\delta q_{\text{sur}} = dU_{\text{sur}} \text{ (constant volume)}$$

$$\delta q_{\text{sur}} = dH_{\text{sur}} \text{ (constant pressure)}$$

Since $dH = dU + d(pV)$, this implies $d(pV)_{\text{sur}} \approx 0$

Entropy of Surroundings

U and H are state functions, so δq_{sur} is independent of reversibility:

$$dS_{\text{sur}} = \frac{\delta q_{\text{sur,rev}}}{T_{\text{sur}}} = \frac{\delta q_{\text{sur}}}{T_{\text{sur}}}$$

As the temperature is invariant,

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T_{\text{sur}}}$$

Work for Irreversible Change

For an irreversible change,

$$|\delta w_{\text{irrev}}| < |\delta w_{\text{rev}}|$$

As $\delta w < 0$,

$$\delta w_{\text{irrev}} > \delta w_{\text{rev}}$$

In general,

$$\delta w \geq \delta w_{\text{rev}}$$

Clausius Inequality

$$\delta w \geq \delta w_{\text{rev}}$$

From the first law of thermodynamics,

$$dU = \delta q + \delta w = \delta q_{\text{rev}} + \delta w_{\text{rev}}$$

which leads to

$$\delta q_{\text{rev}} - \delta q = \delta w - \delta w_{\text{rev}} \geq 0$$

Hence,

$$dS = \frac{\delta q_{\text{rev}}}{T} \geq \frac{\delta q}{T}$$

Isolated System

For an isolated system, $\delta q = 0$

$$dS \geq \frac{\delta q}{T} = 0$$

3rd version of the second law of thermodynamics:

“The entropy of an isolated system never decreases.”

Entropy and Spontaneity

For an isolated system,

$$dS \geq 0$$

- Reversible change: $dS = 0$
 - The change can be reversed
 - Achieved only at equilibrium
- Irreversible change: $dS > 0$
 - The change cannot be reversed
 - Arrow of time: spontaneous change

Fundamental Equation

From the first law,

$$dU = \delta q + \delta w$$

For a reversible change,

$$dU = \delta q_{\text{rev}} + \delta w_{\text{rev}} = TdS - pdV$$

In fact, for both reversible and irreversible changes,

$$dU = TdS - pdV$$

Natural Independent Variables

$$dU = TdS - pdV$$

→ S and V are **natural independent variables** of U

$$U = U(S, V)$$

As U is a state function,

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

Thermodynamics Definitions

$$dU = TdS - pdV$$

As $U = U(S, V)$ is a state function,

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S} \right)_V$$

$$p = - \left(\frac{\partial U}{\partial V} \right)_S$$

Examples

Heat transfer

Expansion (reversible vs. free)

Phase transitions

Heating

Heat Transfer

Transfer of heat $|\delta q|$: h $\rightarrow$ c

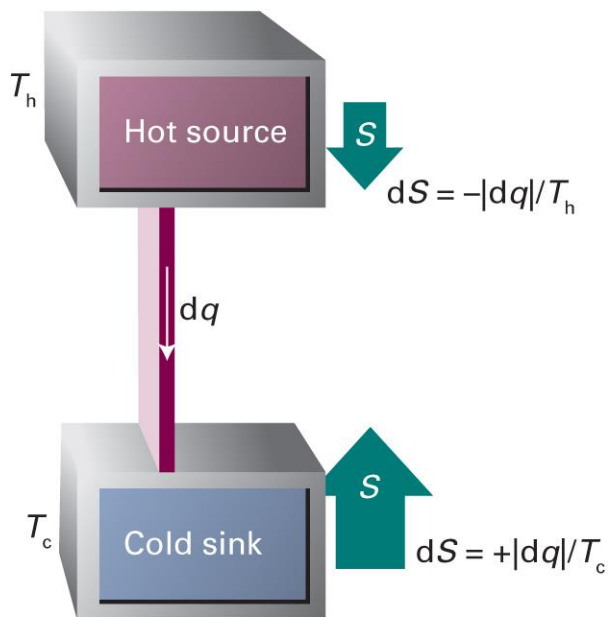
$$dS_h = -|\delta q|/T_h$$

$$dS_c = +|\delta q|/T_c$$

The total entropy change is

$$dS = dS_h + dS_c = |\delta q| \left(\frac{1}{T_c} - \frac{1}{T_h} \right)$$

- Equilibrium: $dS = 0$ or $T_h = T_c$
- Spontaneity: $dS > 0$ or $T_h > T_c$



Reversible Expansion

$$\delta q = -\delta q_{\text{sur}}$$

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = -\frac{q_{\text{rev}}}{T} = -\Delta S$$

Hence,

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} = 0$$

Free Expansion

For a perfect gas, isothermal expansion accompanies with

$$\Delta S = nR \ln \left(\frac{V_f}{V_i} \right)$$

However, $w = 0$ (free expansion) and $\Delta U = 0$ (perfect gas)

$$q = q_{\text{sur}} = 0$$

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = 0$$

Hence,

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} > 0$$

Reversible Phase Transition

Normal transition temperature T_{trs} : the temperature at which two phases are in equilibrium at 1 atm

If the system is at T_{trs} and 1 atm,

$$q = \Delta_{\text{trs}}H \text{ (constant pressure)}$$

$$\Delta_{\text{trs}}S = \Delta_{\text{trs}}H/T_{\text{trs}}$$

$$\Delta_{\text{trs}}S_{\text{sur}} = -\Delta_{\text{trs}}H/T_{\text{trs}}$$

$$\Delta_{\text{trs}}S_{\text{total}} = \Delta_{\text{trs}}S + \Delta_{\text{trs}}S_{\text{sur}} = 0$$

Vaporization Data

	$\Delta_{\text{vap}}H^\ominus / (\text{kJ mol}^{-1})$	$\Delta_{\text{vap}}S^\ominus / (\text{J K}^{-1} \text{mol}^{-1})$	T_b / K
Benzene	30.8	87.2	353
CCl_4	30	85.8	350
Cyclohexane	30.1	85.1	354
H_2S	18.7	87.9	213
Methane	8.18	73.2	112
Water	40.7	109.1	373

Trouton's rule

Trouton's Rule

“A wide range of liquids give approximately the same standard entropy of vaporization (about $85 \text{ J K}^{-1} \text{ mol}^{-1}$)”

liquid $\rightarrow$ gas: a huge change of volume

For a perfect gas, $\Delta S = nR \ln(V_f/V_i)$

Heating at Constant V

$$S(T_f) = S(T_i) + \int_{T_i}^{T_f} \frac{\delta q_{\text{rev}}}{T}$$

At constant volume, $\delta q_{\text{rev}} = dU = C_V dT$

$$S(T_f) = S(T_i) + \int_{T_i}^{T_f} \frac{C_V}{T} dT$$

For constant C_V in $[T_i, T_f]$,

$$S(T_f) = S(T_i) + C_V \int_{T_i}^{T_f} \frac{1}{T} dT = S(T_i) + C_V \ln \left(\frac{T_f}{T_i} \right)$$

Heating at Constant p

$$S(T_f) = S(T_i) + \int_{T_i}^{T_f} \frac{\delta q_{\text{rev}}}{T}$$

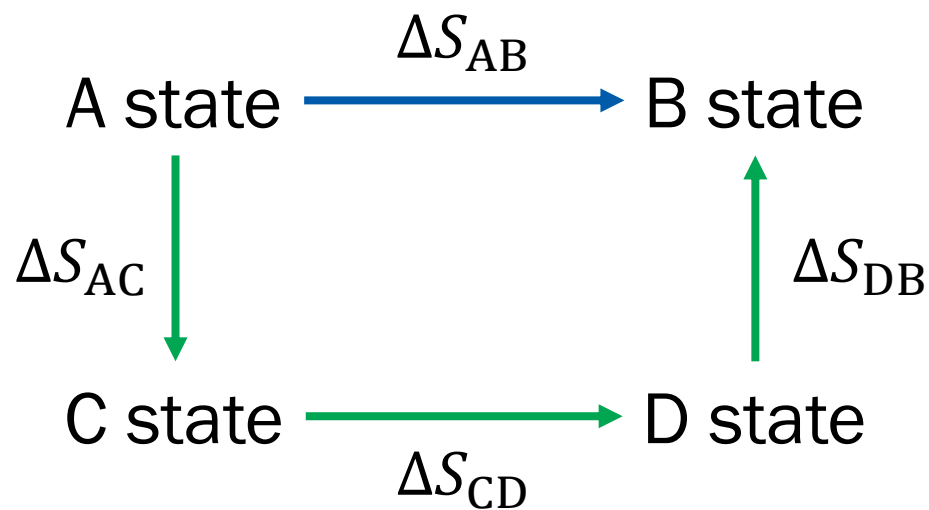
At constant pressure, $\delta q_{\text{rev}} = dH = C_p dT$

$$S(T_f) = S(T_i) + \int_{T_i}^{T_f} \frac{C_p}{T} dT$$

For constant C_p in $[T_i, T_f]$,

$$S(T_f) = S(T_i) + C_p \int_{T_i}^{T_f} \frac{1}{T} dT = S(T_i) + C_p \ln \left(\frac{T_f}{T_i} \right)$$

Entropy as State Function



$$\Delta S_{AB} = \Delta S_{AC} + \Delta S_{CD} + \Delta S_{DB}$$

Example 3B.1

Calculate the entropy change when argon at 25 °C and 1.00 bar in a container of volume 0.500 dm³ is allowed to expand to 1.000 dm³ and is simultaneously heated to 100 °C. (Take the molar heat capacity at constant volume to be $(3/2)R$.)

Stage 1: isothermal expansion at 25 °C, 0.500 dm³ → 1.000 dm³

$$\Delta S(\text{stage 1}) = nR \ln(V_f/V_i)$$

Stage 2: isochoric heating at 1.000 dm³, 25 °C → 100 °C

$$\Delta S(\text{stage 2}) = nC_{V,m} \ln(T_f/T_i) = (3/2)nR \ln(T_f/T_i)$$

Example 3B.1

$$n = pV/RT$$

$$= (1.00 \times 10^5 \text{ Pa}) \times (0.500 \times 10^{-3} \text{ m}^3) / (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) / 298 \text{ K}$$

$$= 0.0201 \text{ mol}$$

$$\Delta S(\text{stage 1}) = nR \ln(V_f/V_i)$$

$$= 0.0201 \text{ mol} \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times \ln(1.000/0.500)$$

$$= 0.116 \text{ J K}^{-1}$$

$$\Delta S(\text{stage 2}) = (3/2)nR \ln(T_f/T_i)$$

$$= (3/2) \times 0.0201 \text{ mol} \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times \ln(373/298)$$

$$= 0.056 \text{ J K}^{-1}$$

Example 3B.1

Therefore,

$$\begin{aligned}\Delta S &= \Delta S(\text{stage 1}) + \Delta S(\text{stage 2}) \\ &= 0.116 \text{ J K}^{-1} + 0.056 \text{ J K}^{-1} = 0.173 \text{ J K}^{-1}\end{aligned}$$

Nernst Heat Theorem

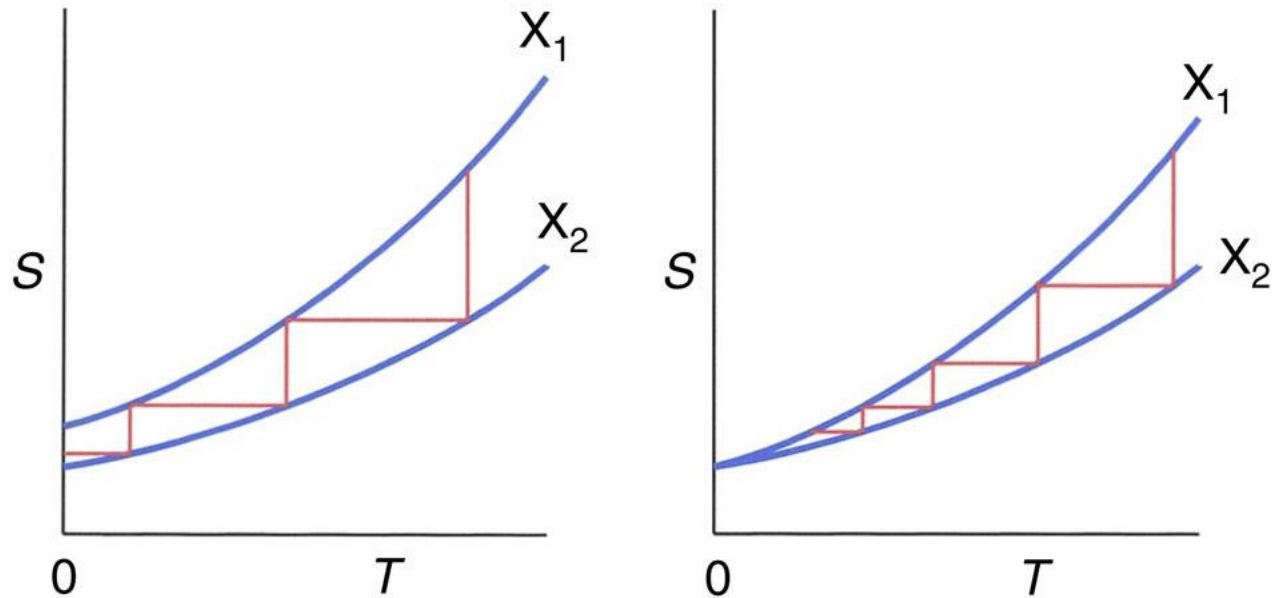
“The entropy change accompanying any transformation approaches zero as the temperature approaches zero:

$$\Delta S \rightarrow 0 \text{ as } T \rightarrow 0,$$

if all the substances involved are perfectly ordered.”

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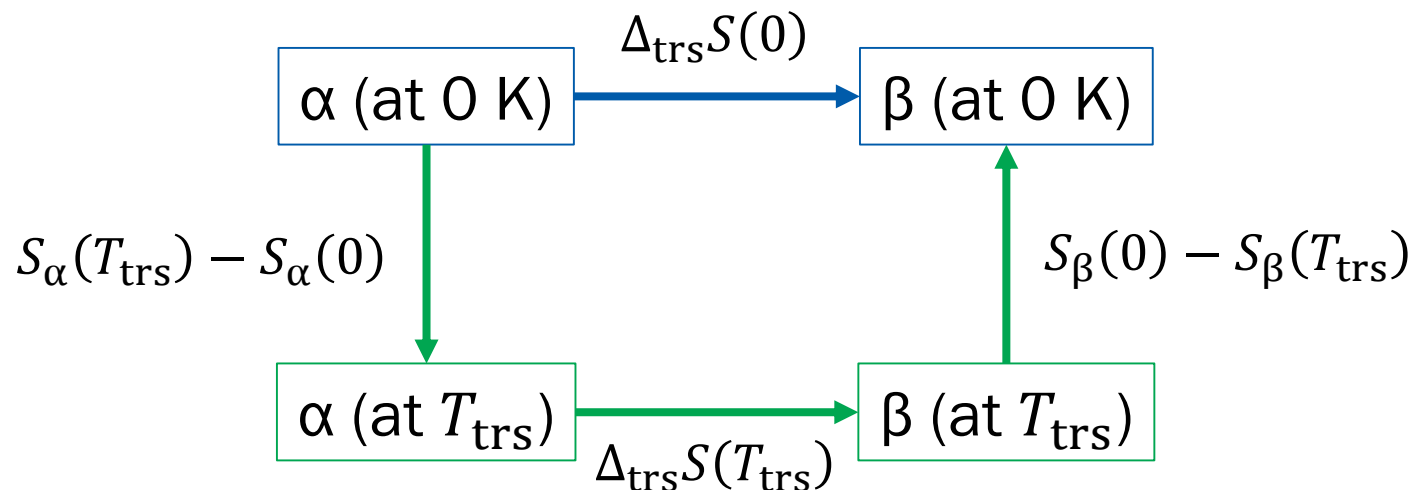
Unattainability Principle



(Image: Masanes and Oppenheim, *Nature Communications* 8: 14538 (2017), Figure 1)

Brief Illustration 3C.2

Calculate the entropy of the transition between orthorhombic sulfur, α , and monoclinic sulfur, β , at 0 K.



Brief Illustration 3C.2

1. α (at 0 K) $\rightarrow$ α (at T_{trs})

$$S_{\alpha,\text{m}}(T_{\text{trs}}) - S_{\alpha,\text{m}}(0) = \int_0^{T_{\text{trs}}} \frac{C_{p,\text{m}}}{T} dT = 37 \text{ J K}^{-1} \text{ mol}^{-1}$$

2. α (at T_{trs}) $\rightarrow$ β (at T_{trs})

$$\Delta_{\text{trs}} S_{\text{m}}(T_{\text{trs}}) = \frac{\Delta_{\text{trs}} H_{\text{m}}(T_{\text{trs}})}{T_{\text{trs}}} = \frac{402 \text{ J mol}^{-1}}{369 \text{ K}} = 1.09 \text{ J K}^{-1} \text{ mol}^{-1}$$

3. β (at T_{trs}) $\rightarrow$ β (at 0 K)

$$S_{\beta,\text{m}}(0) - S_{\beta,\text{m}}(T_{\text{trs}}) = \int_{T_{\text{trs}}}^0 \frac{C_{p,\text{m}}}{T} dT = -38 \text{ J K}^{-1} \text{ mol}^{-1}$$

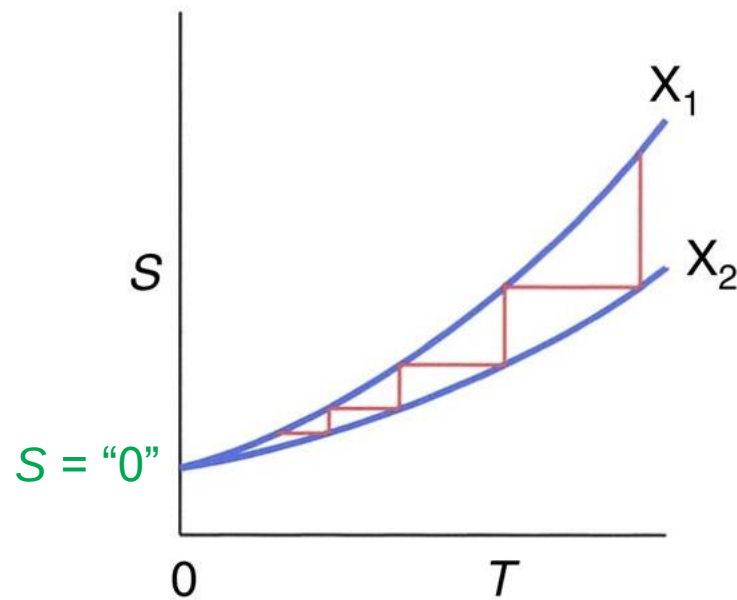
Brief Illustration 3C.2

Therefore,

$$\begin{aligned}\Delta_{\text{trs}}S(0) &= \{S_{\alpha,\text{m}}(T_{\text{trs}}) - S_{\alpha,\text{m}}(0)\} + \Delta_{\text{trs}}S_{\text{m}}(T_{\text{trs}}) + \{S_{\beta,\text{m}}(0) - S_{\beta,\text{m}}(T_{\text{trs}})\} \\ &= 37 \text{ J K}^{-1} \text{ mol}^{-1} + 1 \text{ J K}^{-1} \text{ mol}^{-1} - 38 \text{ J K}^{-1} \text{ mol}^{-1} = 0\end{aligned}$$

Third Law of Thermodynamics

“The entropy of all perfect crystalline substances is zero at $T = 0$.”



(Image: Masanes and Oppenheim, *Nature Communications* 8: 14538 (2017), Figure 1)

Third-Law Entropy

$$S(T) - \underbrace{S(0)}_{= 0} = \int_0^T \frac{C_p(T')}{T'} dT'$$

$$S(T) = \int_0^T \frac{C_p(T')}{T'} dT'$$

Debye T^3 Law

For a crystalline solid at $T \sim 0$,

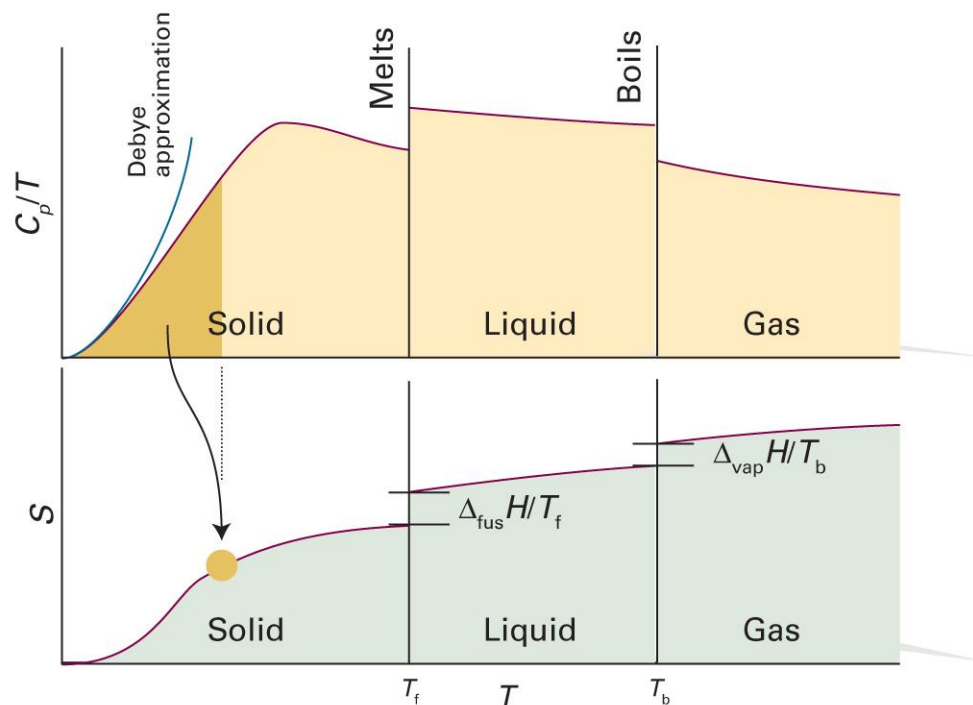
$$C_{p,m}(T) = aT^3$$

Hence,

$$S_m(T) = \int_0^T \frac{C_{p,m}(T')}{T'} dT' = \int_0^T \frac{a(T')^3}{T'} dT' = \frac{1}{3} aT^3 = \frac{C_{p,m}}{3}$$

when T is close to zero.

Typical Entropy Curve



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 3C.1)

Thermochemistry of Entropy

- Standard entropy $S^\ominus$: third-law entropy of a pure substance at 1 bar (always positive)
- Molar standard entropy $S_m^\ominus = S^\ominus/n$
- Standard reaction entropy $\Delta_r S^\ominus$

$$\Delta_r S^\ominus = \sum_{\text{products } i} \nu(i) S_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) S_m^\ominus(j)$$

Enthalpy vs. Entropy

- **Enthalpy:** no natural absolute value

$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) \Delta_f H^\ominus(i) - \sum_{\text{reactants } j} \nu(j) \Delta_f H^\ominus(j)$$

- **Entropy:** third-law absolute value

$$\Delta_r S^\ominus = \sum_{\text{products } i} \nu(i) S_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) S_m^\ominus(j)$$

One Exception

$$S^\ominus(\text{H}^+, \text{aq}) = 0: \text{convention}$$

The entropies of ions in water are values
relative to the hydrogen ion in water
→ can either be positive or negative

Temperature Dependence

$$S_m(T_2) = S_m(T_1) + \int_{T_1}^{T_2} \frac{C_{p,m}(T)}{T} dT$$

Using the definition of standard reaction entropy,

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \int_{T_1}^{T_2} \frac{\Delta_r C_p^\ominus(T)}{T} dT$$

Similar to Kirchhoff's law!

Temperature Dependence

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \int_{T_1}^{T_2} \frac{\Delta_r C_p^\ominus(T)}{T} dT$$

If $\Delta_r C_p^\ominus$ is constant in $[T_1, T_2]$,

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \Delta_r C_p^\ominus \int_{T_1}^{T_2} \frac{1}{T} dT$$

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \Delta_r C_p^\ominus \ln(T_2/T_1)$$

So, what does entropy mean?

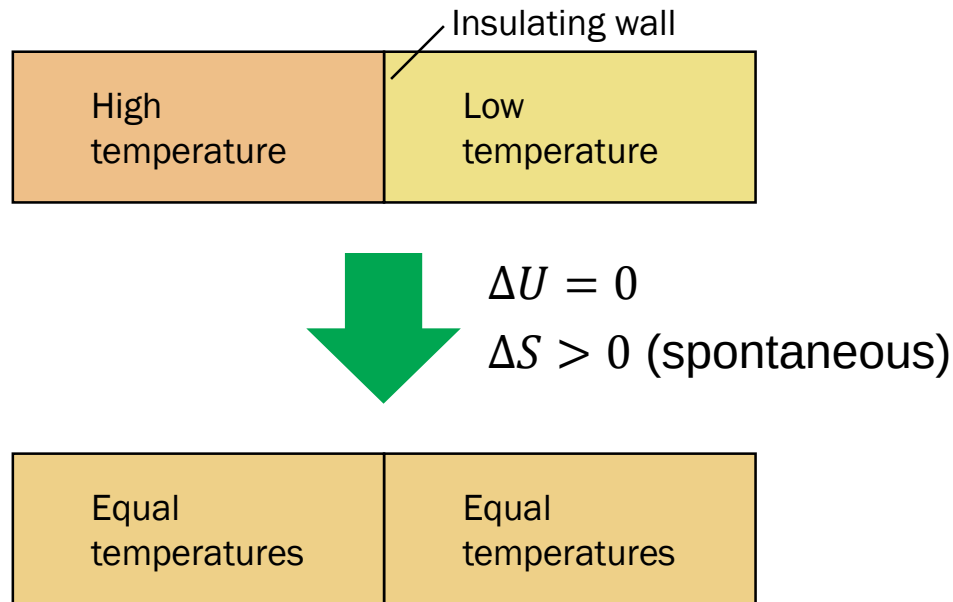
Second Law

$$dS \geq 0 \text{ (for an isolated system)}$$

- $dS = 0$: reversible change, equilibrium
- $dS > 0$: irreversible change, spontaneous change

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Entropy and Work



Loss of capacity to do work

Index of Exhaustion

Entropy can be considered as an **index of exhaustion**

- Clausius: “The energy of the universe is constant; the entropy tends to a maximum.”
- Heat death of the universe: when S_{universe} is maximized
- Jeremy Rifkin, *Entropy: A New World View* (1980)

Statistical Consideration

- (Macro)state: defined by values of **macroscopic** variables
- Microstate: specific **microscopic** arrangement
- Different microstates can be mapped into the same macrostate

Example: Coin Toss

When tossing three coins,

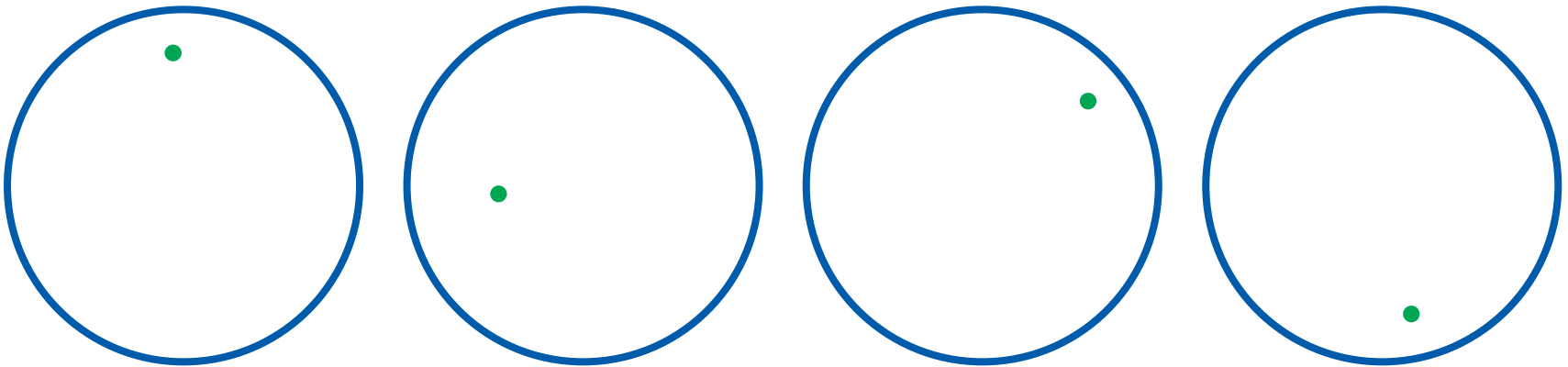
- **Macrostate:** defined by the number of heads h
- **Microstate:** the actual arrangement of coins



Example: Diffusion

When a gas molecule moves around the system,

- **Macrostate:** defined by the volume V
- **Microstate:** the position of the molecule



Number of Microstates W

“The number of microstates
corresponding to a specific macrostate”

- Coin toss

$$W(h = 0) = 1, W(h = 1) = 3, W(h = 2) = 3, W(h = 3) = 1$$

- Diffusion

Let the volume occupied by a molecule v_0 ;

for the system volume V , $W(V) = V/v_0$



Boltzmann's Formula

$$S = k_B \ln W$$

number of microstates

Boltzmann constant (1.38×10^{-23} J/K)

$$\Delta S = S_f - S_i = k_B \ln W_f - k_B \ln W_i = k_B \ln \left(\frac{W_f}{W_i} \right)$$

Example: Diffusion

Suppose that the system volume changes from V_i to V_f :

$$W_i = V_i/v_0$$

$$W_f = V_f/v_0$$

The change in entropy is given by

$$\Delta S_1 = k_B \ln(W_f/W_i) = k_B \ln(V_f/V_i)$$

For a perfect gas consisting of N molecules,

$$\Delta S_N = Nk_B \ln(V_f/V_i) = nR \ln(V_f/V_i)$$

Second Law Revisited

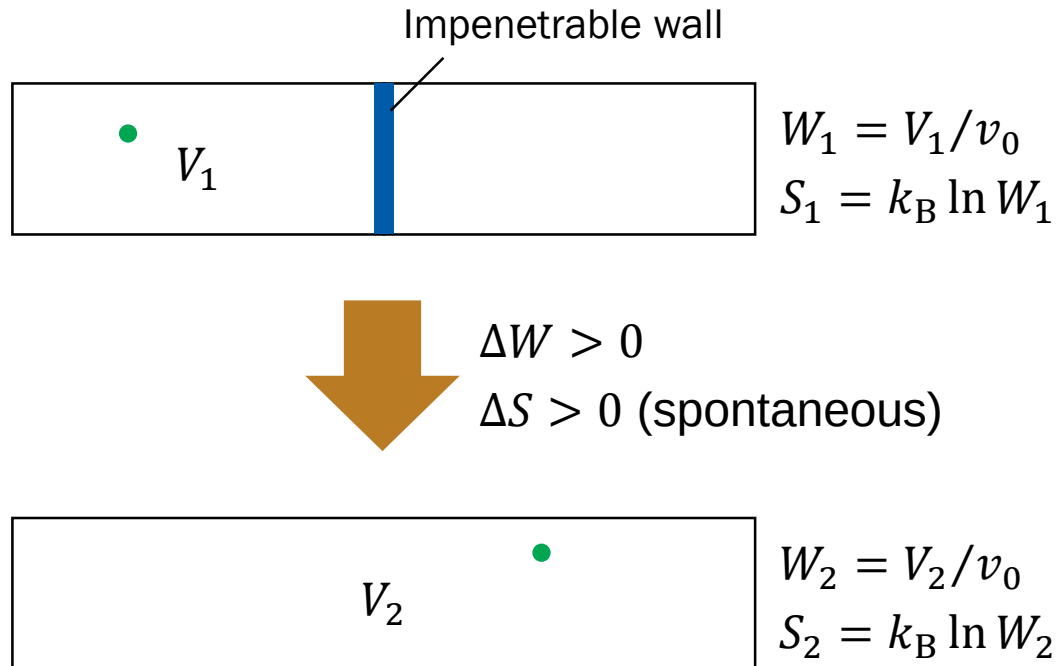
$$dS \geq 0 \text{ (for an isolated system)}$$

As $S = k_B \ln W$ is a monotonic function of W ,

$$dW \geq 0 \text{ (for an isolated system)}$$

The number of microstates never decreases

Example: Diffusion



Third Law Revisited

At $T = 0$,

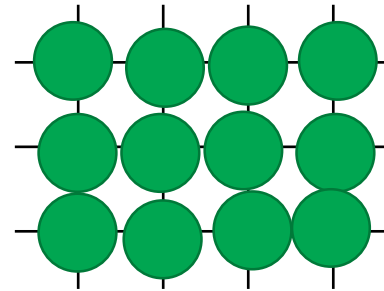
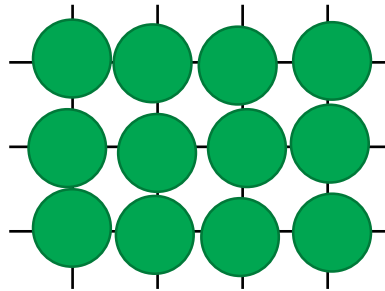
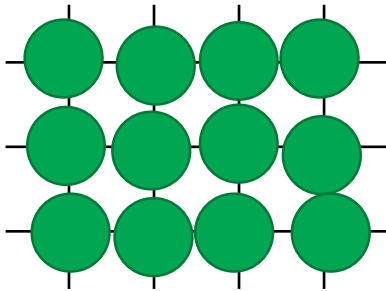
$$S = k_B \ln W = 0$$

$$W = 1$$

Only one microstate is possible at 0 K

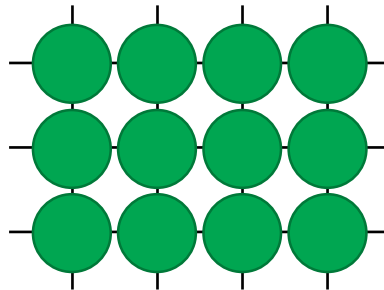
Solid at Absolute Zero

$$T > 0$$



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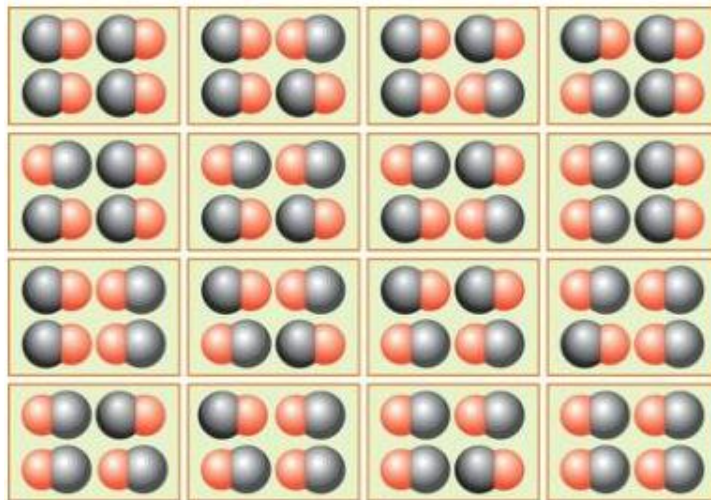
$$T = 0$$



Residual Entropy

$W > 1$ and $S > 0$ even at $T = 0$

e.g. carbon monoxide (CO)



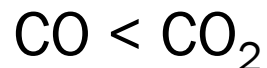
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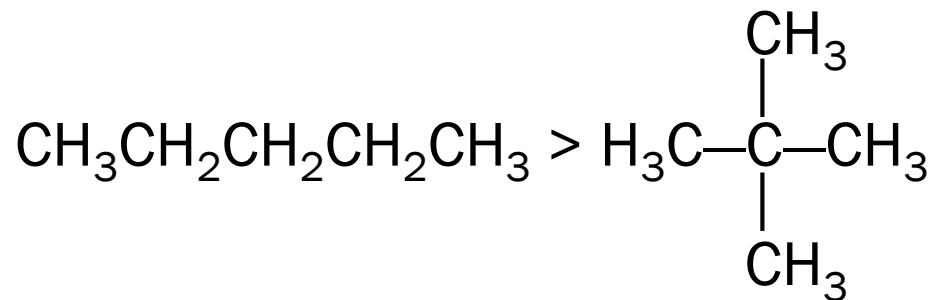
Molecular View on Entropy

In general, W is larger (and S is greater) for a molecule with

1. more atoms



2. more flexibility



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Polymer Conformations



$$\Delta W > 0$$

$$\Delta S > 0 \text{ (spontaneous)}$$



Summary

